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import tensorflow as tf
from tensorflow.keras import layers
import numpy as np
import matplotlib.pyplot as plt

(train_images, _), (_, _) = tf.keras.datasets.mnist.load_data()

train_images = train_images.reshape(-1, 28, 28, 1).astype("float32")
train_images = (train_images - 127.5) / 127.5

BUFFER_SIZE = 60000
BATCH_SIZE = 256

train_dataset = tf.data.Dataset.from_tensor_slices(train_images)
train_dataset = train_dataset.shuffle(BUFFER_SIZE).batch(BATCH_SIZE)

def make_generator_model():
    model = tf.keras.Sequential([
        layers.Dense(7 * 7 * 256, use_bias=False, input_shape=(100,)),
        layers.BatchNormalization(),
        layers.LeakyReLU(),

        layers.Reshape((7, 7, 256)),

        layers.Conv2DTranspose(128, (5, 5), strides=(1, 1),
                                padding='same', use_bias=False),
        layers.BatchNormalization(),
        layers.LeakyReLU(),

        layers.Conv2DTranspose(64, (5, 5), strides=(2, 2),
                                padding='same', use_bias=False),
        layers.BatchNormalization(),
        layers.LeakyReLU(),

        layers.Conv2DTranspose(1, (5, 5), strides=(2, 2),
                                padding='same', use_bias=False,
                                activation='tanh')
    ])
    return model

def make_discriminator_model():
    model = tf.keras.Sequential([
        layers.Conv2D(64, (5, 5), strides=(2, 2),
                        padding='same', input_shape=(28, 28, 1)),
        layers.LeakyReLU(),
        layers.Dropout(0.3),

        layers.Conv2D(128, (5, 5), strides=(2, 2), padding='same'),
        layers.LeakyReLU(),
        layers.Dropout(0.3),
    ])

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        layers.Flatten(),
        layers.Dense(1, activation='sigmoid')
    ])
    return model

generator = make_generator_model()
discriminator = make_discriminator_model()

cross_entropy = tf.keras.losses.BinaryCrossentropy(from_logits=False)

def discriminator_loss(real_output, fake_output):
    real_loss = cross_entropy(tf.ones_like(real_output), real_output)
    fake_loss = cross_entropy(tf.zeros_like(fake_output), fake_output)
    return real_loss + fake_loss

def generator_loss(fake_output):
    return cross_entropy(tf.ones_like(fake_output), fake_output)

generator_optimizer = tf.keras.optimizers.Adam(1e-4)
discriminator_optimizer = tf.keras.optimizers.Adam(1e-4)

@tf.function
def train_step(images):
    noise = tf.random.normal([BATCH_SIZE, 100])

    with tf.GradientTape() as gen_tape, tf.GradientTape() as disc_tape:
        generated_images = generator(noise, training=True)

        real_output = discriminator(images, training=True)
        fake_output = discriminator(generated_images, training=True)

        gen_loss = generator_loss(fake_output)
        disc_loss = discriminator_loss(real_output, fake_output)

    gradients_of_generator = gen_tape.gradient(
        gen_loss, generator.trainable_variables)
    gradients_of_discriminator = disc_tape.gradient(
        disc_loss, discriminator.trainable_variables)

    generator_optimizer.apply_gradients(
        zip(gradients_of_generator, generator.trainable_variables))
    discriminator_optimizer.apply_gradients(
        zip(gradients_of_discriminator,
discriminator.trainable_variables))

def train(dataset, epochs):
    for epoch in range(epochs):
        for image_batch in dataset:

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        train_step(image_batch)

    print(f"Epoch {epoch + 1} completed")

    if (epoch + 1) % 5 == 0:
        generate_and_display_images(generator)

def generate_and_display_images(model):
    noise = tf.random.normal([16, 100])
    predictions = model(noise, training=False)

    plt.figure(figsize=(4, 4))
    for i in range(16):
        plt.subplot(4, 4, i + 1)
        plt.imshow(predictions[i, :, :, 0] * 127.5 + 127.5,
cmap='gray')
        plt.axis('off')
    plt.show()

train(train_dataset, epochs=5)

plt.figure(figsize=(4, 4))
for i in range(16):
    plt.subplot(4, 4, i + 1)
    plt.imshow(train_images[i, :, :, 0] * 127.5 + 127.5, cmap='gray')
    plt.axis('off')
plt.suptitle("Real MNIST Images")
plt.show()

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generate_and_display_images(generator)
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Downloading data from <https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz>

11490434/11490434 ————— 0s 0us/step

/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:93: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

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    super().__init__(activity_regularizer=activity_regularizer,
**kwargs)

```

/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

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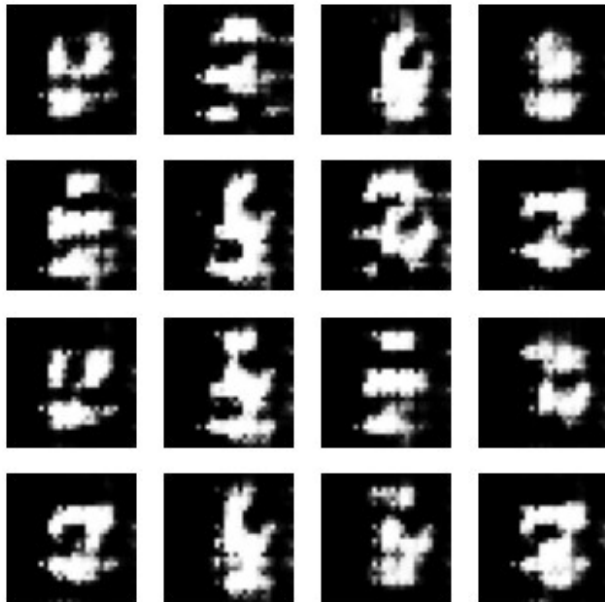
    super().__init__(activity_regularizer=activity_regularizer,
**kwargs)

```

Epoch 1 completed

Epoch 2 completed

Epoch 3 completed
Epoch 4 completed
Epoch 5 completed



Real MNIST Images

